

JORDAN NEWMAN

Backend / Infrastructure Engineer | Reliability, Networking, Security

Marlboro, NJ | 347-739-4731 | phoem@mac.com | Private GitHub repository access available upon request

SUMMARY

Backend and infrastructure engineer specializing in high-performance systems, networking, security, reliability, automation, and large-scale production operations. Built and operated infrastructure spanning thousands of servers and developed custom software for traffic analysis, DDoS mitigation, authentication, telemetry, deployment automation, and high-traffic web serving.

TECHNICAL SKILLS

Languages: C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic

Systems: FreeBSD, Linux, kernel modules, system calls, bootloaders, operating systems, kqueue, GDB, TCP/IP

Networking/Security: DDoS detection/mitigation, packet capture, DNS analysis, libpcap, IPFW, authentication, telemetry

Embedded: Embedded C, Atmel AVR, SPI, UART, I2C, Arduino, circuit design, soldering, hardware troubleshooting

Cloud/DevOps: Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation

Databases/Web: MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Built backend and infrastructure systems supporting thousands of servers, 10 CDN locations, and 65+ Gbps peak traffic.
- Developed high-performance C web server, packet analysis tools, DDoS mitigation systems, and authentication platforms.
- Automated deployment, provisioning, monitoring, telemetry, and infrastructure management across production environments.
- Integrated enterprise ERP, OAuth, SMTP OAuth, SFTP, supplier catalogs, and MultiValue/Universe database workflows.
- Implemented hands-on Docker, Kubernetes, Terraform, Azure DevOps, CI/CD, and security-scanning workflows.

SELECTED SYSTEMS PROJECTS

TAFOS Operating System: Developed educational x86 OS kernel in C and assembly with custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

KeepClean: Developed FreeBSD kernel module hooking syscalls to monitor execution activity, protect system assets, and report telemetry to userspace services.

VirtualDir: Developed FreeBSD kernel module remapping filesystem paths through syscall interception; deployed customer-facing across approximately 30 production servers over 5-8 years.

StatCache: Developed FreeBSD kernel module instrumenting stat() and lstat() activity for filesystem visibility and performance analysis.

PrimeDump: Developed libpcap-based traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 headers with ncurses real-time views and IPFW integration.

PrimeDNSTop: Developed DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source/domain activity.

AVR Smart Smoke/CO2 Detector: Developed contract firmware in embedded C for battery-powered wireless Atmel AVR prototypes using SPI, UART, and I2C interfaces.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist | Remote / Tampa Bay, FL | July 2022 - Present

- Implemented Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.
- Partnered with architecture teams to standardize deployment practices, DevOps policies, and production software delivery.
- Developed internal AI workflows, reusable libraries, and documentation to improve engineering productivity and consistency.

DDI Systems - Senior Development Specialist | Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET software by porting components to C#, improving maintainability and long-term reliability.
- Integrated Microsoft Office 365 SMTP OAuth, replacing deprecated authentication with secure customer-facing workflows.

ISPrime LLC - CEO | Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational cleanup, and software platform improvements for production hosting infrastructure.
- Refined internally developed monitoring, security, and infrastructure tools to improve reliability and operational efficiency.

DDI Systems - Computer Programmer | Manalapan, NJ | April 2017 - April 2018

- Integrated ERP software with SellerCloud and Essendant systems, automating order, catalog, image, and SDS data workflows.
- Developed customer-facing import tools for MultiValue/Universe databases and SFTP-based supplier integrations.

Too Much Media LLC - Computer Programmer | Morganville, NJ | January 2016 - April 2017

- Developed real-time browser chat integrated with ticketing systems, improving customer support responsiveness.
- Implemented OAuth 2.0 authorization for protected websites and customer-facing access workflows.

MFCXY, Inc. - Computer Programmer | Chicago, IL | December 2014 - September 2015

- Developed backend server features and Windows client integrations while identifying performance and security issues.
- Built middleware to migrate data across database platforms and improve operational continuity.

AJPM, LLC - Programmer / Owner | New Jersey / Remote | November 2012 - December 2013

- Built automated purchasing and inventory software tracking acquisition, repair, resale, and financial workflows.

ISPrime Inc. - CIO / Partner | *Weehawken, NJ* | *March 2001 - December 2014*

- Built and operated infrastructure spanning 3,000-4,000 servers across 10 CDN locations and multiple datacenters.
- Sustained peak traffic exceeding 65 Gbps while supporting mission-critical customer workloads.
- Developed PrimeHTTPD, a high-performance C web server optimized around FreeBSD kqueue supporting 150,000+ concurrent connections.
- Deployed PrimeHTTPD across approximately 200 production servers supporting high-traffic customer environments.
- Developed DDoS detection and mitigation systems protecting infrastructure during attacks involving up to 750,000 compromised hosts.
- Created PrimeDump and PrimeDNSTop, libpcap-based network analysis tools for traffic anomalies and DNS attacks.
- Developed production FreeBSD kernel modules including KeepClean, VirtualDir, and StatCache for security and filesystem observability.
- Built monitoring, telemetry, authentication, and security systems used across thousands of production servers.

FatWallet.com - Systems Administrator | *Wisconsin / Remote* | *December 2000 - June 2001*

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime.
- Built automated alerting systems for website availability, server health, and operational response.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.